

Bitwise linear trail through **ARADI** found by SAT

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1 ARADI

Maximal linear correlation: 2^{-1}

8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0
ARADI - random																																	
8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2		
8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2		
PseudoR																																	
8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2		
8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2		

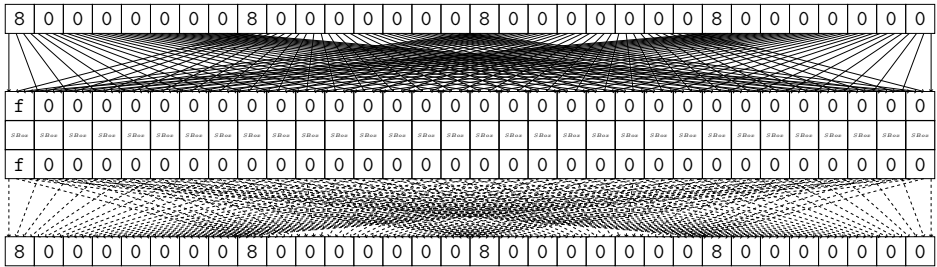
2 ARADI-round0

Maximal linear correlation: $2^{-1.0}$

8	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	
8	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	
BN																														
8	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	
8	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	
SBias Layer																														
8	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	
8	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	
Linear Layer0																														
8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	2
8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	2

3 SBoxLayer

Maximal linear correlation: $2^{-1.0}$



4 LinearLayer0

Maximal linear correlation: $2^{-0.0}$

8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0
L0								L0								L0								L0															
8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	2	
8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	2	